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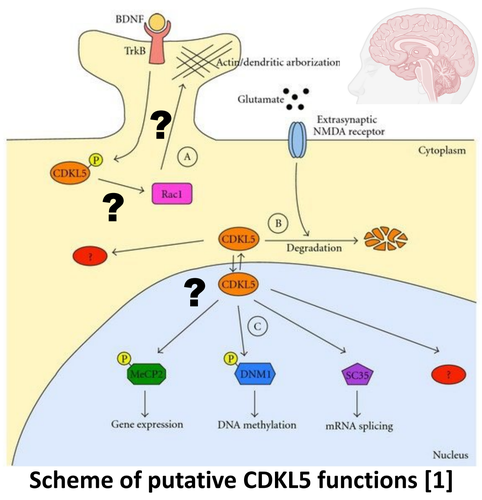
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SMASH machine learning for science and humanities postdoctoral program
I FEEL SLOVENIA
Co-funded by the European Union

A pipeline for shortlisting & lead optimizing *de novo* designed protein binders towards CDKL5

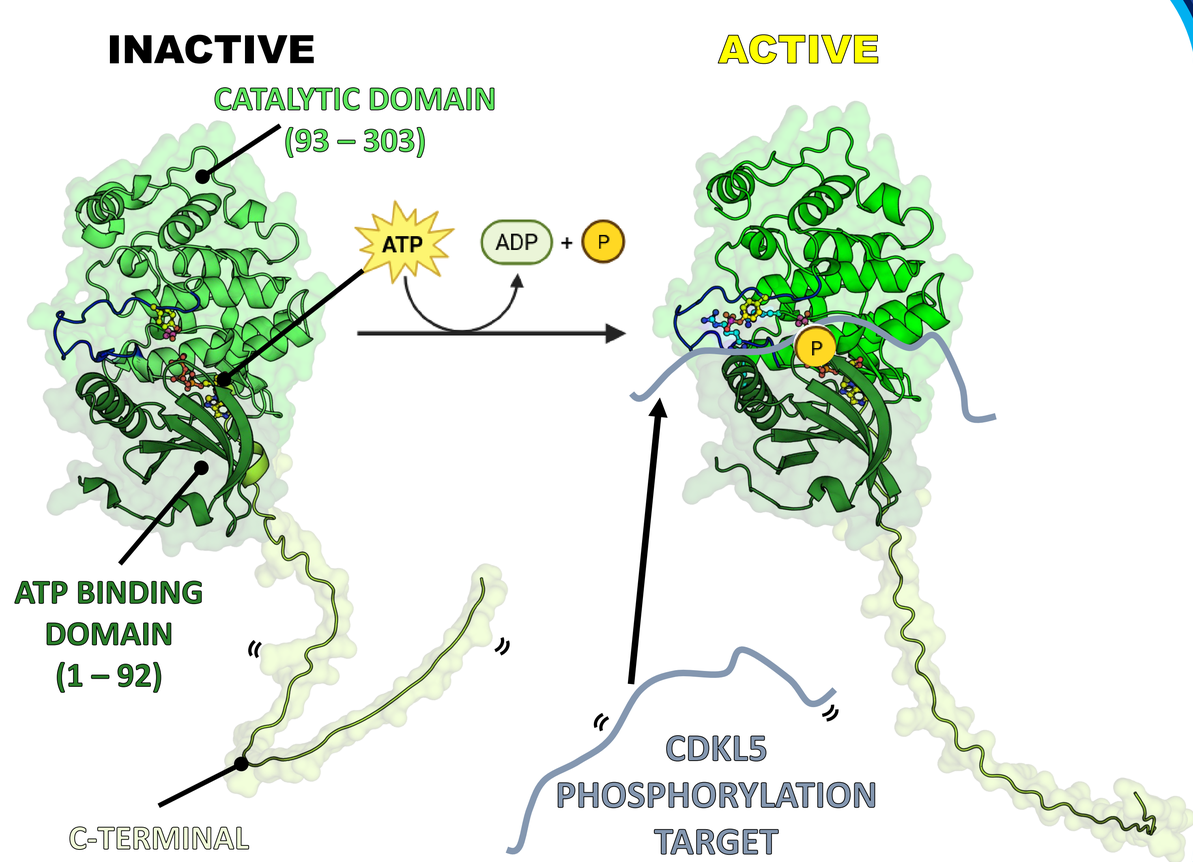
BACKGROUND ON CDKL5

CDKL5 (cyclin-dependent kinase-like 5, Uniprot ID: O76039) is an enzyme that activates upon auto-phosphorylation and then mediates the phosphorylation of microtubule binding proteins and transcription factors essential for crucial brain development [1]. Putative alternative splicing isoforms might exist, with a specific yet unknown region-temporal expression pattern, while non-synonymous genetic mutations can cause CDKL5 Deficiency Disorder, leading to severe neurodevelopmental disabilities from early infancy [2]. To advance our understanding of CDKL5 biology in normal and disease conditions, it is necessary to develop reagents to *in vivo* visualize and quantify the relative amounts of the different isoforms. Traditional IgG antibodies, typically used for specific molecular recognition, have proven difficult to be obtained due to the high conservation and low immunogenicity of the folded part.

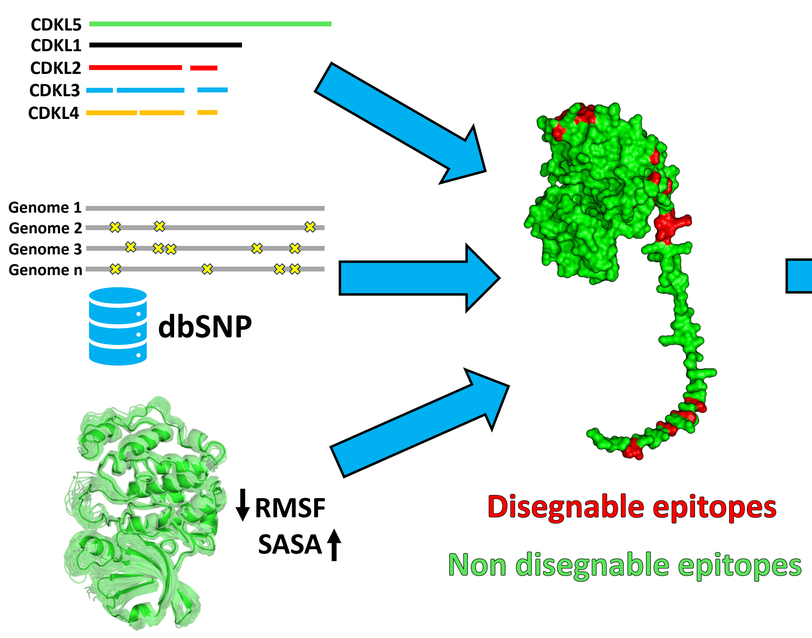


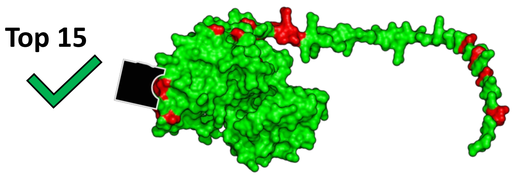
Scheme of putative CDKL5 functions [1]

The C-terminal tail of CDKL5 can be used to develop antibodies, but there is the need to get a larger coverage of the target epitopes for differentiating among the isoforms. Machine Learning *de novo* design methods are promising for making protein binders selective for specific antigen epitopes [3]. Indeed, such methods may represent one of the few available options when working with poorly immunogenic target molecules, unfolded epitopes and highly conserved antigens. Anyway, despite the great acceleration in developing new methods in the last couple of years [4-5], >> 10 designs have still to be characterized to obtain binders with low nM affinity and decent to good developability properties (stable, soluble, non immunogenic and specific for your target). Indeed, CDKL5 has poor solubility for setting a high-throughput *in vitro* screening. During my project I will set and use a multi-step workflow for finding ready-to-apply CDKL5 binders, while minimizing the experimental validation work.



HIGH THROUGHPUT *IN SILICO DE NOVO* BINDER DESIGN + SCORING



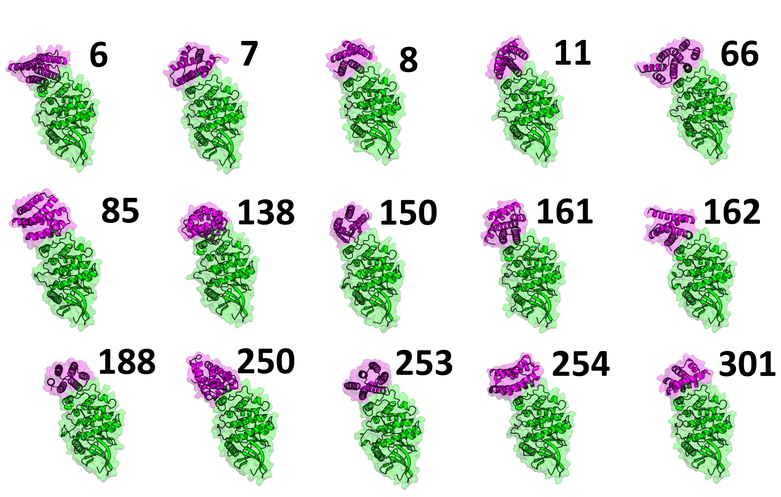


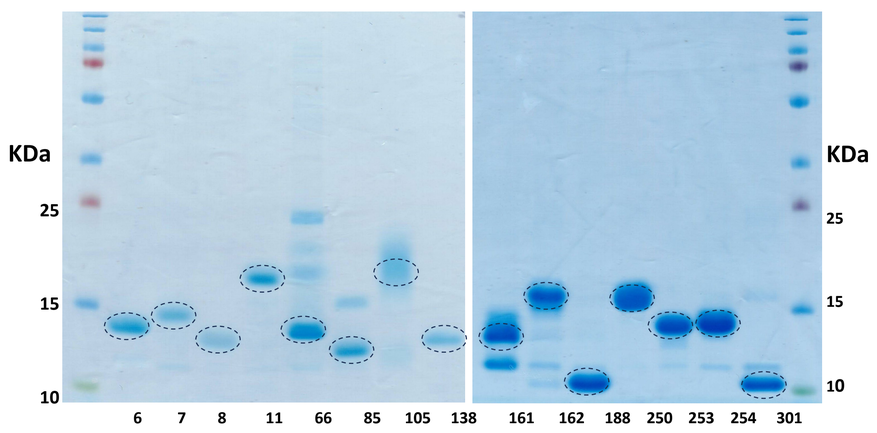
Monomer refolding filters
- Av. plddt > 90 & min > 70 (ESMFold)
- pTM > 0.9 (ESMFold)

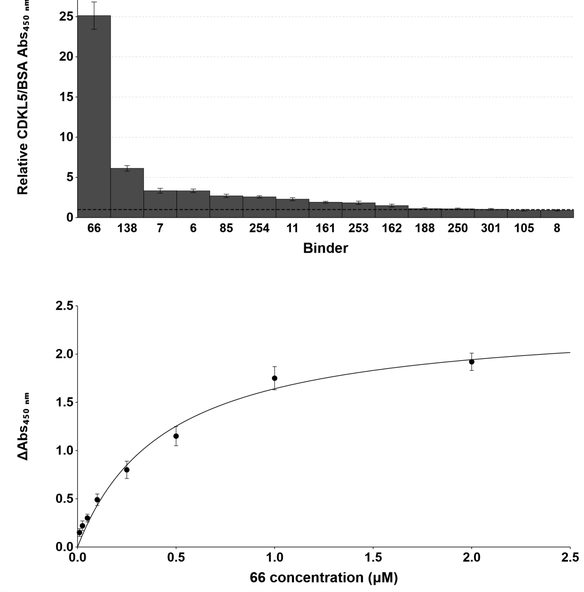
Complex refolding filters
- RosettaDDG/ΔSASA < -20
- Deep learning co-folding < 2 Å
- Rigid-body re-docking < 2 Å
- AI-enhanced MD-simulations binder RMSD with frames superposed along target structure

Selection criteria
Top-15 ipSAE (Boltz-2/AF3)
Structural diversity

TOP 15 DESIGNS VALIDATION

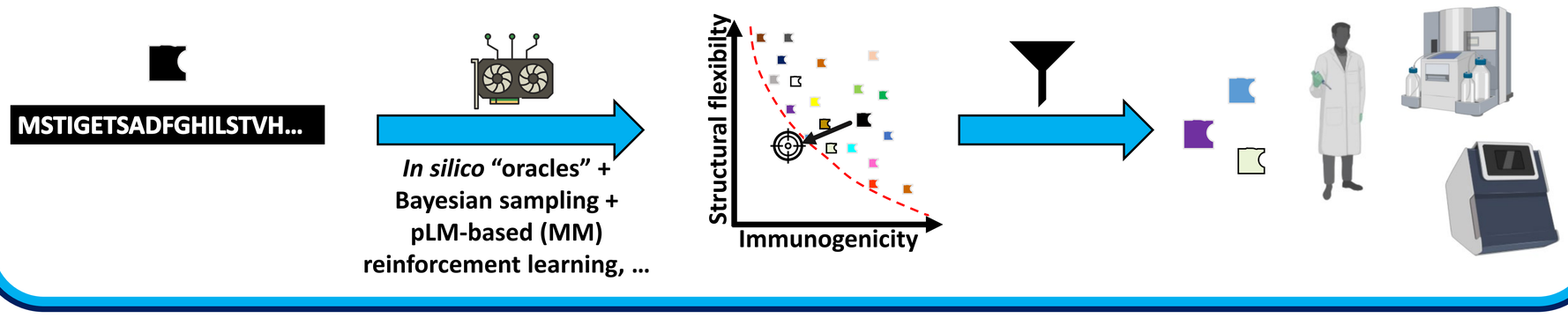






ELISA assay indicates there is at least one good candidate (66)

FUTURE GOAL: MULTI-MODAL (MM) LEAD OPTIMIZATION OF BEST DESIGN(S)



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